# Setting text in columns

*A line of text is easiest to read when it holds about sixty to seventy characters. On a wide page that means wide margins or a large font. Columns are the other answer: they split the page into measures of a comfortable length.*

## Why columns

Newspapers, journals and magazines have set their text in columns for as long as there have been large pages. A single line across an A4 page in a ten point font holds well over a hundred characters, and the eye easily loses its place on the way back to the start of the next line. Two columns halve that distance and keep the type size that a reader is used to.

Columns also make better use of the page. Short paragraphs, lists and tables that would leave much of a wide line empty fill a column nicely, and a page with two columns holds more text than a page with one wide measure and generous margins.

## What makes a column

> *A column is a page within the page.*

A column is a page within the page. Text flows into the first column until it is full, then on into the second one, and only after the last column of a page on to the next page. Everything a page break does applies at a column break too: a heading stays with the paragraph that follows it, a paragraph keeps at least two lines together at the top and at the bottom, and a table repeats its header row.

The width of a column follows from the width of the page, the number of columns and the gap between them. On this page, with 174 mm of content width, two columns and a gap of 7 mm, each column is 83.5 mm wide.

| Columns | Gap | Width | Characters |
|---|---|---|---|
| 1 | none | 174 mm | about 125 |
| 2 | 7 mm | 83.5 mm | about 60 |
| 3 | 6 mm | 54 mm | about 38 |

## Narrow measures

A narrow column has fewer spaces per line to take up the leftover width when the text is justified. A line breaker that looks at a whole paragraph at once finds better breaks than one that fills each line as far as it can, and hyphenation adds breakpoints where a long word would otherwise leave a loose line.

The narrower the column, the more these two matter. Three columns on an A4 page are about as narrow as continuous text should get; below that, the spaces in a justified line grow visibly wide, and ragged right text becomes the better choice.

## Floats in a column

A box with a side float stays in its column, and the lines of the column run around it, as the quotation above shows. A float that goes to the top or the bottom of the page goes to the top or the bottom of its column instead, at the width of the column. Footnotes do the same: each column collects the notes of its own text.[1]

Running heads, page numbers and the other contents of the page margins belong to the page, not to a column. They are placed once per page, as in a document with one column.

## Breaking between columns

Automatic breaks move the text on to the next column whenever the current one is full. A forced break with `break-before: column` starts an element at the top of the next column; on the last column of a page, that is the first column of the next page. A forced page break, `break-before: page`, skips the remaining columns and starts a new page right away.

Outside a multi-column layout a column break has no effect, so the same stylesheet works for a document with one column.

## Balancing

On the last page, the columns are balanced: the text is spread so that they end at about the same height, instead of filling the first column and leaving the second one short. The same happens above an element that spans the columns, such as the title of this article, which is set with `column-span: all`. With `column-fill: auto` the last page fills its columns one after the other.

1. This note belongs to the second column and is set at its foot, as wide as the column.